NUTRITIONAL GUIDANCE – PREDICTIVE HEALTH INTERVENTION SYSTEM

Project Id: 25-26J-351

Project Proposal Report

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August 2025

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

..... 27 / 08 / 2025

Signature of the supervisor Date

ABSTRACT

This research introduces an AI-powered Nutritional Guidance and Predictive Health Intervention System designed to transform conventional dietary tracking into a proactive, personalized nutrition management platform. The system integrates machine learning, circadian rhythm analysis, and real-time health monitoring to deliver intelligent meal recommendations, predict potential micronutrient deficiencies up to three weeks in advance, and adapt suggestions to individual lifestyles, biometrics, and goals.

The solution collects and processes user data such as age, weight, allergies, sleep patterns, dietary preferences, and ongoing medications. Using predictive analytics, it identifies nutritional gaps and aligns meal suggestions with the body's natural rhythms through a circadian nutrition engine. A medication–food interaction alert module enhances safety by notifying users of potential conflicts, while a goal-based recommendation system dynamically optimizes meals for targets such as weight management, muscle gain, or energy enhancement.

User engagement is strengthened through an interactive feedback loop that tracks progress, provides motivational nudges, and adapts recommendations based on behavioral changes. The system's backend architecture unifies AI subsystems with a responsive dashboard, offering nutrient intake tracking, upcoming risk alerts, and real-time adaptation. By shifting from reactive tracking to predictive intervention, this research delivers a clinically aware, adaptive, and user-centric nutrition solution that promotes long-term health, safety, and sustainable dietary habits.

Keywords: *Predictive nutrition, AI-powered health intervention, circadian nutrition, micro-nutrient deficiency prediction, personalized dietary recommendations*

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LIST OF ABBREVIATIONS

- **AI** - Artificial Intelligence
- **API** - Application Programming Interface
- **B2B** - Business-to-Business
- **IDEs** - Integrated Development Environments
- **LMM** - Large Multimodal Model
- **ML** - Machine Learning
- **UI** - User Interface

1. INTRODUCTION

Figure 1 – Nutrition Guidance – Predictive Health Intervention System

The AI-Powered Kitchen Ecosystem is a comprehensive research initiative designed to address the global challenges of food waste and poor nutrition through an integrated, AI-driven platform. This holistic system seamlessly combines four specialized modules: a Spontaneous Cooking Assistant that uses advanced computer vision to identify available ingredients and suggest relevant recipes, an AI Shopping Agent that provides voice-enabled, intelligent product search and cross-platform price comparisons, a Behavioral Food Expiry Predictor that learns from user habits to deliver personalized spoilage alerts, and a Nutritional Guidance and Predictive Health Intervention System that provides AI-powered, personalized dietary and wellness recommendations. Together, these components create a synergistic workflow that supports the entire food lifecycle, from planning and purchasing to cooking and consumption, fostering sustainable household practices and reducing waste [1], [2], [25].

Within this integrated framework, my contribution, the Nutritional Guidance and Predictive Health Intervention System, acts as the intelligent core dedicated to user wellness. This module transcends traditional nutrition apps by leveraging machine learning to proactively forecast micronutrient deficiencies weeks in advance, aligning meal recommendations with the user's circadian rhythms for optimal metabolic health, and integrating critical medication-food interaction alerts to ensure safety [5], [6], [26]. By synthesizing data from other ecosystem components, such as consumed recipes and grocery purchases. It delivers deeply personalized, adaptive, and clinically-informed dietary guidance, transforming the platform from a mere kitchen tool into a proactive health management companion [22], [23], [27].

1.1 Background & Literature Survey

Nutrition is a fundamental determinant of human health, influencing disease prevention, cognitive performance, and overall quality of life. However, maintaining a balanced diet remains a challenge for many individuals due to busy lifestyles, lack of nutritional awareness, and inconsistent dietary tracking. Poor dietary habits can result in micronutrient deficiencies, chronic diseases, and reliance on highly processed foods [1].

Traditional nutrition management tools, such as MyFitnessPal and Cronometer, primarily focus on calorie tracking and static meal planning without addressing real-time lifestyle variations or predictive nutritional needs. Chi and Demiris [2] explored early concepts of nutrition-aware cooking in smart kitchens, while Zhang and Wang [3] demonstrated large multimodal models (LMMs) for food recognition and assistance. More recent AI-driven solutions have applied machine learning (ML) for personalized meal suggestions [4], but most remain reactive, identifying deficiencies only after they occur.

Emerging research has emphasized the role of circadian nutrition, which aligns food intake with biological rhythms to optimize metabolism and energy levels [5]. Additionally, integrating medication–food interaction alerts has been identified as an important safety measure, especially for individuals on long-term prescriptions [6]. Despite these developments, there is limited adoption of systems that unify predictive analytics, biological timing, and adaptive personalization into a single comprehensive nutritional guidance framework.

1.2 Research Gap

Although AI-powered dietary tools have advanced, several gaps remain:

- Lack of predictive capabilities to identify micro-nutrient deficiencies before they manifest [19], [20]
- Limited integration of circadian rhythm–based meal recommendations into mainstream nutrition systems [5], [21]
- Minimal consideration for medication–food interaction alerts in consumer-grade dietary platforms [6], [16]
- Insufficient adaptive personalization based on ongoing lifestyle, biometric, and behavioral changes [22], [23]
- Fragmentation between nutritional analysis, goal setting, and long-term habit-building support [17], [24]

This absence of a unified, clinically aware, and adaptive nutrition platform creates a gap in delivering proactive dietary guidance to individuals seeking sustainable health improvement.

1.3 Research Problem

Current nutrition management systems are largely reactive, focusing on tracking past dietary intake rather than anticipating future nutritional needs [1]–[4]. They often fail to integrate circadian rhythm–based recommendations [5], [7], overlook potential food–medication interactions [6], and provide static guidance that does not adapt to changes in an individual's lifestyle, biometric data, or behavioral patterns [7], [8]. This fragmented approach limits the effectiveness of dietary management, preventing individuals from achieving optimal health outcomes and sustainable habit formation [1]–[6]. Therefore, there is a critical need for a unified, predictive, and adaptive nutrition management system that delivers personalized, real-time dietary guidance while considering biological rhythms, medication interactions, and ongoing lifestyle changes [7], [8].

1.4 Comparison with Existing Systems

A number of digital nutrition platforms currently dominate the market, such as MyFitnessPal and Cronometer, which primarily provide calorie counting and basic nutrient tracking functionalities [2], [4]. However, these systems operate largely on a reactive model, focusing on historical data logging rather than forward-looking, predictive health intervention. Studies have shown that existing applications often lack intelligent features and are limited in their ability to provide truly personalized guidance [15]. Emerging AI-driven solutions have begun incorporating machine learning for meal suggestions [4], but they still fall short in several critical areas essential for holistic health management, such as safety and adaptability [16].

Feature	Existing System	Proposed System
Core Functionality	Reactive Tracking and logging	Proactive, Predictive Intervention [19], [27], [28]
Deficiency Awareness	Identifies current intake gaps	Predictive deficiencies 2 weeks in advance [19], [27]
Meal Timing	Not Considered	Circadian Rhythm-Based Scheduling [5], [30]
Medicine Safety	Not available	Integrated Medication Food Interaction Alerts [6], [26]
Adaption Level	Static	Dynamic [22], [23], [29]
Primary Goal	Calorie Tracking	Holistic Health Intervention and Habit Formation [17], [28], [29]

Table 1 -- Comparison with Existing System

As illustrated in Table 1, conventional systems are effective for retrospective dietary logging but lack capabilities for anticipatory guidance [1]–[4]. Their absence of circadian-based meal timing [5], [7], medication safety features [16], and adaptive personalization based on continuous user feedback represents a significant limitation, often leading to low long-term user adherence [17]. This proposed system is engineered specifically to address these gaps by integrating predictive analytics, chrononutrition, and safety modules, thereby shifting the paradigm from passive nutritional tracking to active, intelligent, and preventative health management. This comparative advantage directly addresses the deficiencies identified in the research gap (Section 1.2) and establishes the foundation for tackling the research problem outlined in Section 1.4.

2. OBJECTIVES

2.1 Main Objectives

- Develop an AI-driven system that provides personalized nutritional recommendations based on user biometrics, lifestyle, dietary preferences, and goals.
- Implement predictive analytics for proactive health intervention, forecasting potential micro-nutrient deficiencies, and enabling preventive dietary planning.

2.2 Specific Objectives

- Collect and process user data, including age, weight, dietary preferences, allergies, sleep patterns, and activity levels for personalized nutrition profiling.
- Track nutrient intake through meal logging, recipe scanning, and automated nutrient analysis.
- Design and train machine learning models that predict micro-nutrient deficiencies 2 weeks in advance using dietary and biometric data.
- Implement a circadian nutrition engine that aligns meal suggestions with the user's biological clock, work schedule, and sleep patterns.
- Provide goal-oriented meal recommendations, such as high-protein meals for muscle recovery or energy-boosting options for fitness activities.
- Integrate a medication-food interaction alert system to ensure safe consumption and prevent adverse health effects.
- Develop an interactive feedback module offering motivational nudges, habit tracking, and adaptive guidance to encourage long-term behavior change.
- Design a responsive and user-friendly dashboard for nutrient tracking, risk alerts, and personalized meal suggestions.
- Ensure scalable data management and system performance for handling multiple user profiles and large nutritional datasets.

3. METHODOLOGY

3.1 Overall System Architecture

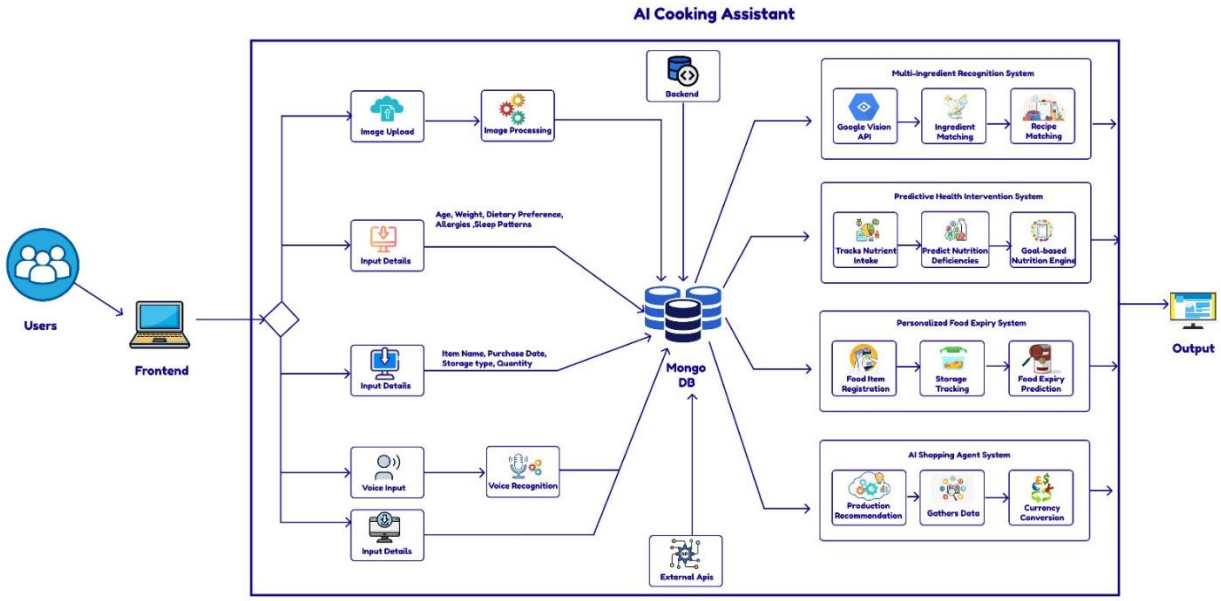

Figure 2 – Overall System Architecture

3.1.1 Nutritional Guidance System Diagram

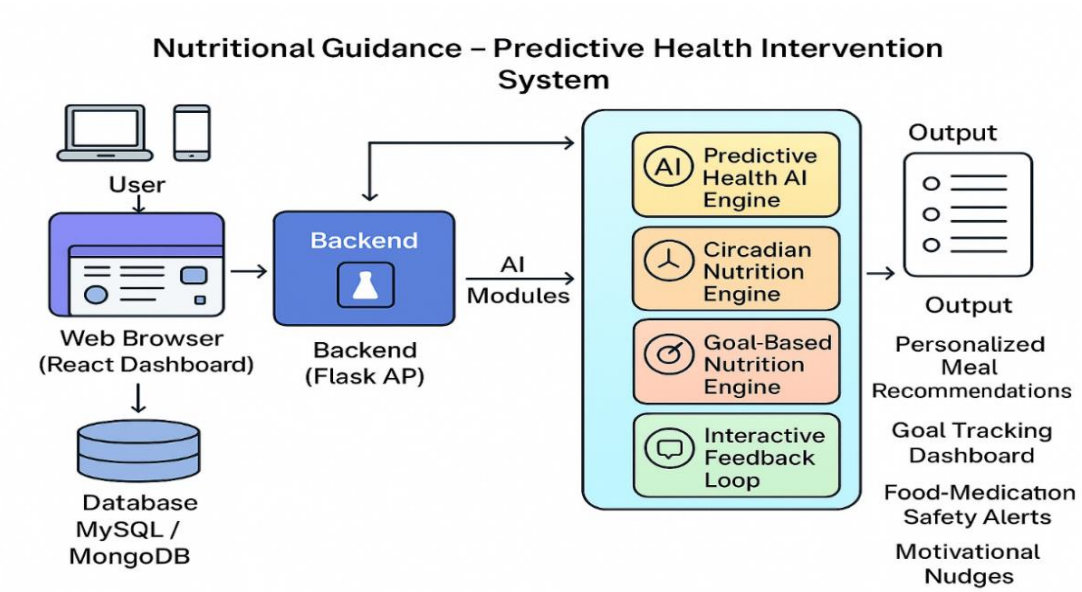

Figure 3 – Nutritional Guidance System Architecture

3.2 Tools and Technologies

- VS Code to implement the code
- Python Libraries for AI & ML
- Scikit-learn / TensorFlow for micro-nutrient deficiency prediction
- Pandas & NumPy for data analysis and preprocessing
- Flask for backend API integration
- React for responsive web dashboard
- Food & Nutrition APIs (USDA / Edamam) for nutrient data retrieval [18]
- MySQL / MongoDB for user profiles, dietary logs, and nutrient records
- Testing Tools: Selenium, PyTest, JMeter for automated and performance testing
- Cloud Services for scalable deployment
- GitHub / GitLab for version control and collaborative development

3.3 Commercialization

The "Nutritional Guidance – Predictive Health Intervention System" is an AI-powered digital health solution designed to revolutionize personalized nutrition and preventive healthcare. By integrating predictive analytics, circadian nutrition, and real-time health monitoring, the system provides proactive dietary interventions and intelligent recommendations. The commercialization of this platform focuses on making advanced nutrition management accessible to individuals, healthcare providers, and wellness organizations worldwide [9], [10].

3.3.1 Commercialization Strategy:

- **Subscription-Based Platform**: Offer the system as a mobile and web-based subscription service, where users can access personalized meal recommendations, nutrient predictions, and food-medication safety alerts [10].

- **B2B Partnerships**: Collaborate with hospitals, nutritionists, fitness centers, and corporate wellness programs to integrate the system into professional healthcare and employee wellness packages [11].
- **API & Data Services**: License AI modules and nutrition analytics APIs to third-party health applications, enabling integration into existing healthcare ecosystems.
- **Target Customers**: The primary clients include health-conscious individuals, patients with chronic conditions, dieticians, fitness enthusiasts, and wellness organizations.
- **System Accessibility**: Accessible through web dashboards and mobile applications, ensuring usability across devices and remote access [12].

3.3.2 Future Scope

- Wearable Integration**:** Integrate with smartwatches and fitness trackers to provide real-time feedback on dietary needs, activity levels, and circadian rhythm alignment [13].
- AI-Powered Health Coach**:** Extend the system with a conversational AI assistant that offers personalized coaching, motivational nudges, and adaptive guidance [11].
- Personalized Supplement Guidance: Recommend supplements (vitamins, minerals) based on predictive nutrient deficiency analysis [14].
- Multi-Language & Cultural Adaptation**:** Expand food databases to include diverse cuisines, making the system globally adaptable.
- Telemedicine Integration: Provide healthcare professionals with dashboards to remotely monitor patient nutrition and progress [12].
- Predictive Disease Risk Analysis**:** Enhance predictive models to assess long-term risks for chronic diseases such as diabetes, obesity, and cardiovascular disorders [14].

3.4 Budget and Budget Justification

For this research component, we do not aim to commercialize our individual components and the whole research will be commercialized as an application.
Following is the budget justification for the whole research system.

Description	Cost (LKR)
Electricity	5000.00
Internet	5000.00
Communication	2000.00
Transportation	3000.00
Paper Publishing	15000.00
Backend Hosting	12000.00
Software Purchasing	10000.00
Prototype Materials (Basic Devices)	8000.00
Total	**60000.00**

Table 2 -- Budget Table

3.4.1 Budget Justification

1. **Electricity (LKR 5,000):** Covers power consumption during system development, AI model training, and testing.
2. **Internet (LKR 5,000):** High-speed internet for research, cloud services, and API access.
3. **Communication (LKR 2,000):** Costs for phone calls, mobile data, and collaboration platform subscriptions to ensure seamless coordination among the research team
4. **Transportation (LKR 3,000):** Travel for research meetings, presentations, or data collection if needed.
5. **Paper Publishing (LKR 15,000):** Costs for conference or journal publication of the research findings.

6. **Backend Hosting (LKR 12,000):** Cloud services for hosting the system and ensuring scalable access.
7. **Software Purchasing (LKR 10,000):** Licenses for necessary software, IDEs, databases, and AI libraries not available for free.

8. **Prototype Materials (LKR 8,000):** Funds for basic hardware components (e.g., sensors, microcontrollers) to build and test a physical proof-of-concept.

4. PROJECT REQUIREMENTS

4.1 Functional requirements

- **User Data Collection:** Receive and store user information, including age, weight, dietary preferences, allergies, sleep patterns, and fitness goals.

- **Dietary Logging:** Allow users to log meals manually or by uploading recipes for nutrient tracking.

- **Personalized Recommendations:** Recommend meals and nutrition plans based on user profile, goals, and predicted deficiencies.

- **Predictive Analytics:** Forecast potential micro-nutrient deficiencies using AI/ML models.

- **Circadian Nutrition Engine:** Suggest meal timings aligned with user's biological clock, work schedule, and sleep patterns.

- **Medication-Food Interaction Alerts:** Show a page where users select a medication (e.g., Panadol) and see foods to avoid, with warnings and alternatives.

- **Interactive Feedback and Habit Tracking:** Provide motivational nudges, progress tracking, and adaptive guidance to support healthy habits.

- **Dashboard and Visualization:** Display nutrient intake, predicted risks, and daily meal suggestions in an intuitive UI.

- **Integration with APIs:** Connect with nutrition databases and food APIs

4.2 Non-functional requirements

- **Fast Response:** Provide quick processing of user data and generation of recommendations.
- **Accurate and Personalized Guidance:** Deliver highly accurate and personalized nutritional recommendations.
- **User-Friendly Design:** Simple, clean, and intuitive dashboard for users of all ages.
- **Availability:** System operates 24/7 without downtime.
- **Security:** Protect all user data against unauthorized access, misuse, or breaches.
- **Cross-Device Compatibility:** Works seamlessly on web across different browsers.
- **Scalability:** Capable of handling a growing number of users and large data volumes.
- **Maintainability:** Modular and well-documented system for easy updates, AI model improvements, and bug fixes.
- **Engagement:** Keep users motivated through alerts, progress tracking, and feedback.

5. GANNT CHART

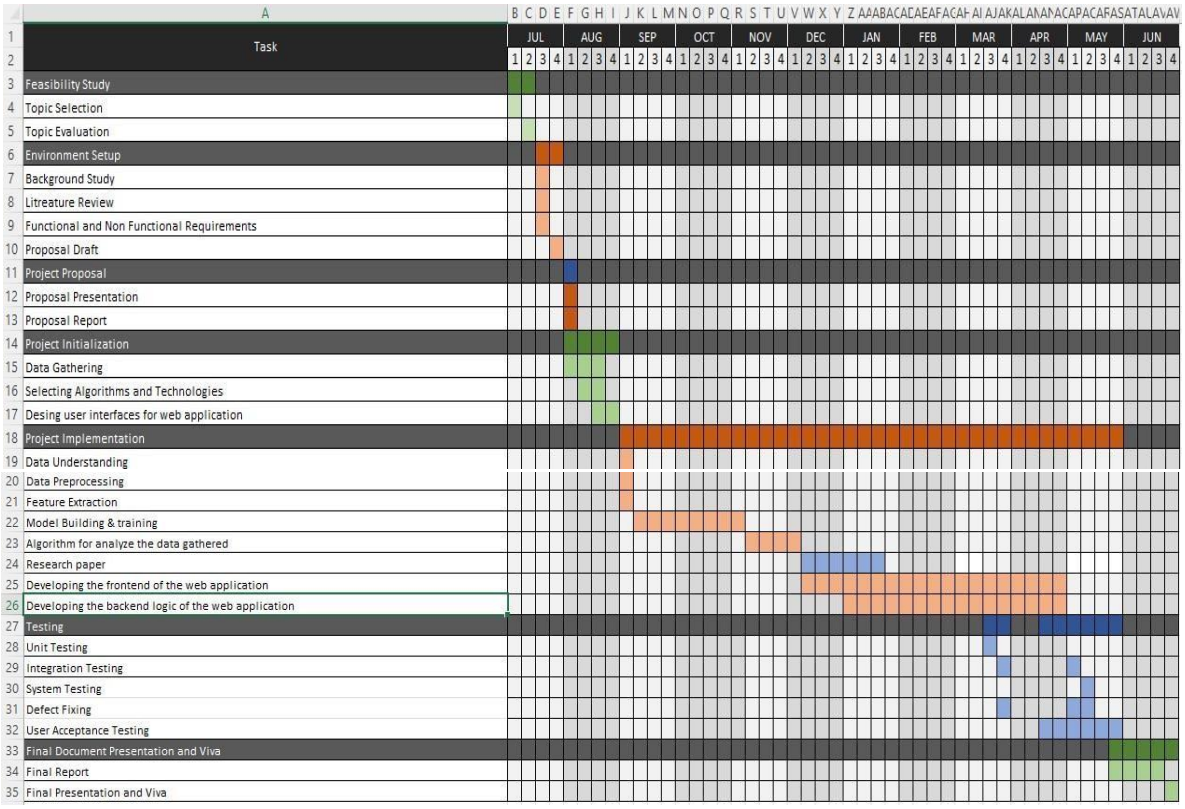

Figure 4 -- Gannt Chart

6. WORK BREAKDOWN STRUCTURE

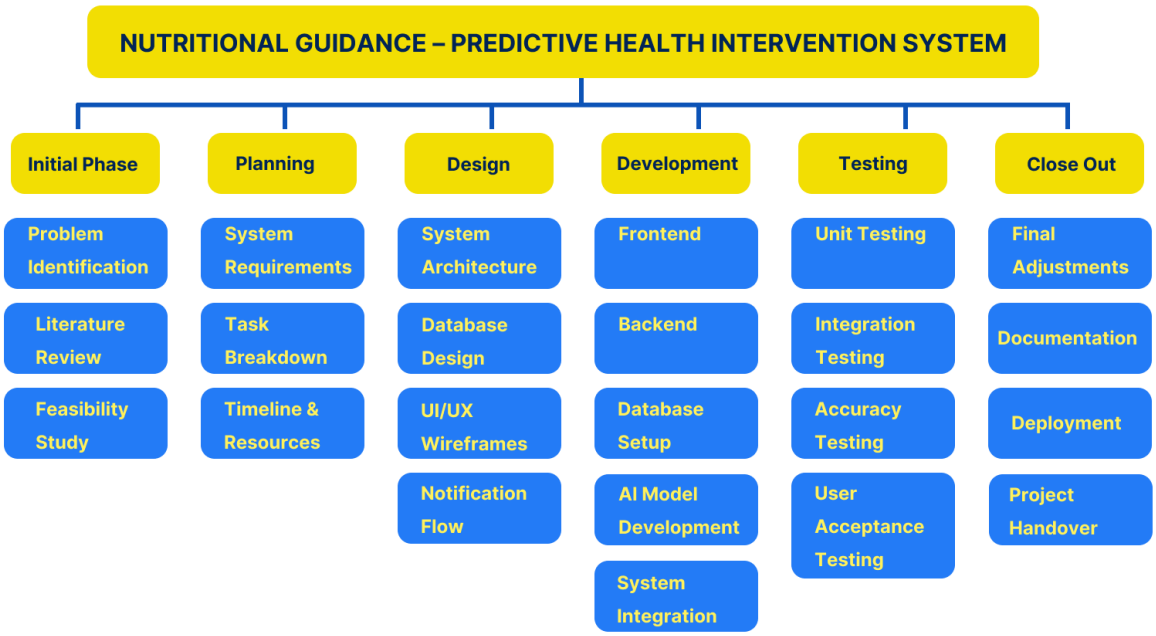

Figure 5 -- Work Breakdown Chart

7. DESCRIPTION OF PERSONNEL AND FACILITIES

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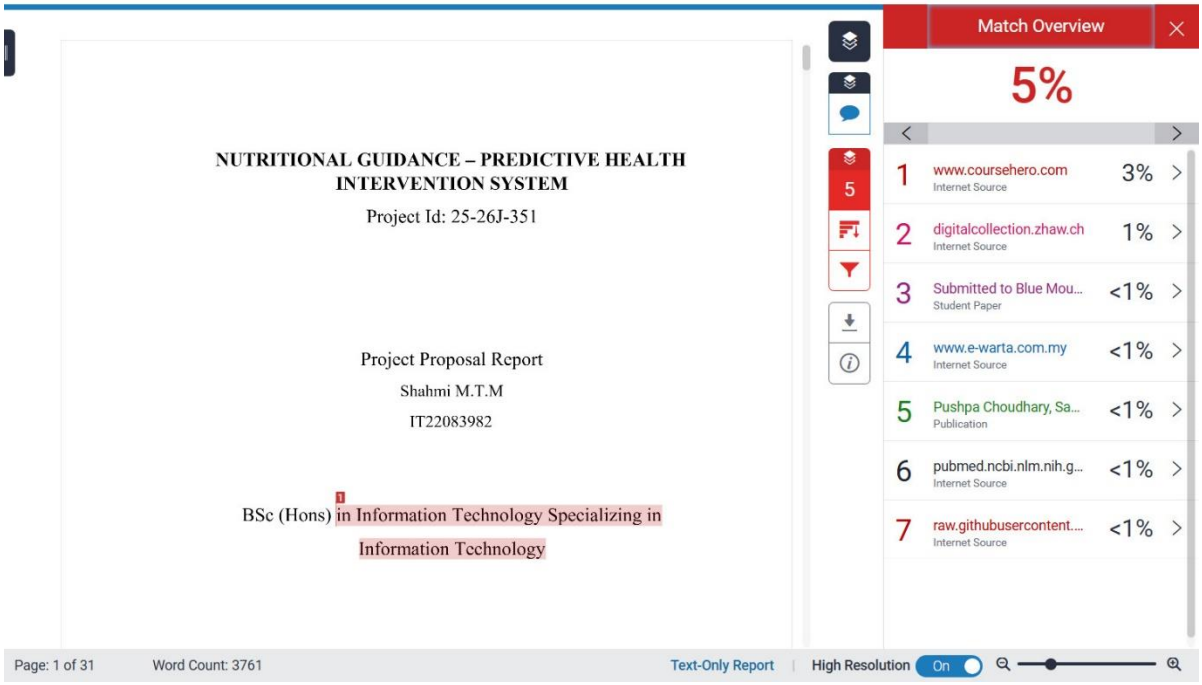
9. APPENDICES

Appendix - A: Plagiarism Report



Appendix 1 -- Plagiarism Report